

Probability and Probability Distributions

$6^{(5)}$

Binomial Probability

Distributions (contd...)

Binomial Distributions

DEFINITION OF BINOMIAL DISTRIBUTION

If X is a discrete random variable with probability mass function, then

$$P(X) = {}^nC_r p^r (1 - p)^{n - r},$$

where $r = 0, 1, 2, 3, \dots, n$ and $q = 1 - p$, then X is a binomial variate and the distribution of X is called binomial distribution.

Binomial Distributions

- The word “binomial” literally means “two numbers.” A binomial distribution for a random variable X (known as binomial variate) is one in which there are only two possible outcomes, success and failure, for a finite number of trials.

Binomial Distributions

- Note that however we define success and failure, the two events must be mutually exclusive and complementary, that is, they cannot occur at the same time (mutually exclusive), and the sum of their probabilities is 100%(complementary)

Binomial Distributions

combination of x success from n trials

Number of success

Number of failures

$$P(X = r) = {}^nC_r p^r (1 - p)^{(n - r)},$$

Probability of success

Random variable X

probability of failure

Binomial Distributions

- Where $p + q = 1$ (because probability cannot be greater than 1)
- X is a discrete random variable which may take only countable number of distinct values such as 1, 2, 3, 4, . . .

Binomial Distributions

Assumptions of Binomial Distribution

- ❑ For each trial there are only two possible outcomes on each trial, S (success) & F (failure).
- ❑ The number of trials “ n ” is finite
- ❑ For each trial the two outcomes are mutually exclusive.

Binomial Distributions

- $P(S) = p$ is a constant, $P(F) = q = 1 - p$
- The trials are independent, the outcomes of a trials is not affected by the outcome of any other trial.
- The probability of success, p is constant from trial to trial.

Binomial Distributions

Example 3. A quiz consists of 10 multiple choice questions. Each question has 5 possible answers, only one of which is correct. Petros plans to guess the answer to each question. Find the probability that Petros gets

- a. one answer correct
- b. all 10 answers correct

Binomial Distributions

Solution 3: Given $n = 10$ and $p = 0.2$

a. $P(X = r) = {}^nC_r p^r (1 - p)^{n - r},$

$$\begin{aligned} P(X = 1) &= {}^{10}C_1 (0.2)^1 (1 - 0.2)^{10 - 1} \\ &= \frac{10!}{1!(10 - 1)!} \times (0.2) \times (0.8)^9, \\ &= 10 \times (0.2) \times (0.8)^9 = 0.2684 = 26.84\% \end{aligned}$$

Binomial Distributions

b. $P(X = r) = {}^nC_r p^r (1 - p)^{n - r},$

$$P(X = 10) = {}^{10}C_{10} (0.2)^{10} (1 - 0.2)^{10 - 10},$$

$$= \frac{10!}{10!(10 - 10)!} \times (0.2)^{10} \times (0.8)^0,$$

$$= 1 \times (0.2)^{10} \times 1 = 0.00000001$$

Binomial Distributions

Example 4. The probability that a baby is born a boy is 0.51. A midwife delivers 10 babies. Find

- a. the probability that exactly 4 are boys.
- b. the probability that at least 8 are boys.

Binomial Distributions

Solution 4: Given $n = 10$ and $p = 0.51$

$$\text{a. } P(X) = {}^nC_x p^x (1 - p)^{n - x},$$

$$\begin{aligned} P(X = 4) &= {}^{10}C_4 (0.51)^4 (1 - 0.51)^{10 - 4}, \\ &= \frac{10!}{4!(10 - 4)!} \times (0.51)^4 \times (0.49)^6, \\ &= 0.197 \end{aligned}$$

Binomial Distributions

$$\begin{aligned}P(X \geq 8) &= P(X = 8) + P(X = 9) + P(X = 10) \\&= {}_{10}C_8(0.51)^8(0.49)^2 + {}_{10}C_9(0.51)^9(0.49)^1 \\&\quad + {}_{10}C_{10}(0.51)^{10}(0.49)^0 \\&= 0.04945 + 0.01144 + 0.00119 \\&= 0.0621\end{aligned}$$

Binomial Distributions

Example 5:- In a college 20% of students are girls. In a random sample of 5 students, find the probability that there are at most 2 girls.

Binomial Distributions

Solution 5:- Given $n = 5$ and $p = 0.2$

$$\begin{aligned} P(X \leq 2) &= p(x = 0) + p(x = 1) + p(x = 2) \\ &= {}^5C_0(0.2)^0(0.8)^5 + {}^5C_1(0.2)^1(0.8)^4 \\ &\quad + {}^5C_2(0.2)^2(0.8)^3 \\ &= 0.32768 + 0.4096 + 0.2048 \\ &= 0.94208 \end{aligned}$$

Binomial Distributions

Example 6:- A factory produces electronic chips. Each chip has a 3% chance of being defective. If 20 chips are inspected, find the probability that at most one chip is defective.

Binomial Distributions

Solution 6:- Given $n = 20$ and $p = 0.03$

$$\begin{aligned} P(X \leq 1) &= p(x = 0) + p(x = 1) \\ &= {}^{20}C_0 (0.03)^0 (0.97)^{20} + {}^{20}C_1 (0.03)^1 (0.97)^{19} \\ &= 0.544 + 20(0.03 \times 0.561) \\ &= 0.544 + 0.326 \\ &= 0.8702 \end{aligned}$$

Binomial Distributions

Example 7:- In a statistics exam 45% of students pass. If a class of 20 students takes the exam, find the probability that

- i. exactly 15 students pass
- ii. at least two students pass
- iii. more than 16 students pass

Binomial Distributions

Solution 7:- i. Exactly 15 students pass

$$\begin{aligned}P(X = 15) &= {}^{20}C_{15}(0.45)^{15}(0.55)^5 \\&= 15504 \times 0.000006283 \times 0.050329438 \\&= 0.004902575 \approx 0.0049\end{aligned}$$

ii. at least two students pass

$$P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$$

Binomial Distributions

$$= 1 - [P(X = 0) + P(X = 1)]$$

$$= 1 - [20 C_0 (0.55)^{20} + 20 C_1 \times (0.45) \times (0.55)^{19}]$$

$$= 1 - [(0.55)^{20} + 20 \times (0.45) \times (0.55)^{19}]$$

$$= 1 - 0.0001$$

$$= 0.9999$$

Binomial Distributions

iii. more than 16 students

$$\begin{aligned} P(X > 16) &= P(X = 17) + P(X = 18) + P(X = 19) + P(X = 20) \\ &= {}^{20}C_{17}(0.45)^{17}(0.55)^3 + {}^{20}C_{18}(0.45)^{18}(0.55)^2 \\ &\quad + {}^{20}C_{19}(0.45)^{19}(0.55)^1 + {}^{20}C_{20}(0.45)^{20}(0.55)^0 \\ &= 1140(0.45)^{17}(0.55)^3 + 190(0.45)^{18}(0.55)^2 \\ &\quad + 20(0.45)^{19}(0.55)^1 + (0.45)^{20} \\ &= 0.0003 \end{aligned}$$

Binomial Distributions

CONCLUSION.

- The binomial distribution is a discrete probability distribution used where there are only two possible outcomes for a random variable: *success* and *failure*.

For example if you purchase a lottery ticket, you're either going to win money, or you aren't.

Binomial Distributions

Basically, anything you can think of that can only be a *success* or a *failure* can be represented by a binomial distribution.

☐ Success and failure are mutually exclusive.

They cannot occur at the same time.

Binomial Distributions

- For the binomial distribution to be applied, each successive trial must be *independent* of the last; that is, the outcome of a previous trial has no bearing on the probabilities of success on subsequent trials.

Binomial Distributions

- The binomial distribution assumes a finite number of trials, n .
- A machine is known to produce, on average, 2% defective components. As an engineer you might need to know the probability that 3 items are defective in the next **20** produced.

Binomial Distributions

Therefore the binomial distribution is used when:

- There is a *fixed number of trials (n)*.
- Each trial has *only two outcomes*: Success or Failure.
- The *probability of success (p)* is the same for each trial.
- Trials are *independent*.

Any Questions



Any Questions



Thank
you